

Qno. 1:

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2$$

$$\alpha = 0.05$$

$$n_1 = 16$$

$$\bar{X}_1 = \frac{\sum X_{1i}}{n_1} = 9897.5$$

$$s_1^2 = \frac{\sum (X_{1i} - \bar{X}_1)^2}{n_1 - 1} = 62005060$$

$$n_2 = 12$$

$$\bar{X}_2 = 4120.83$$

$$s_2^2 = 6147944.7$$

Degrees of Freedom:

$$v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{(s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)}$$
$$= \frac{(3875316.25 + 512328.725)^2}{(3875316.25)^2/15 + (512328.725)^2/11}$$
$$= 18.78$$

$$v \approx 19$$

Critical region: $t < -t_{0.025, 19}$ or $t > t_{0.025, 19}$

$$t < -2.093 \text{ or } t > 2.093$$

$$t = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}$$

$$= \frac{5776.67}{2094.67}$$

$$t = 2.758$$

As, $2.758 > 2.093$, we reject H_0 .

Thus, we can say that there is a significant difference between both averages.

Qno.2:

$$H_0: \sigma_A^2 = \sigma_B^2$$

$$H_1: \sigma_A^2 \neq \sigma_B^2$$

$$\alpha = 0.10$$

$$n_A = 9$$

$$\bar{X}_A = 6.71/9 = 0.7456$$

$$S_A^2 = 0.01083$$

$$n_B = 9$$

$$\bar{X}_B = 6.04/9 = 0.6711$$

$$S_B^2 = 0.01249$$

$$f = \frac{S_B^2}{S_A^2}$$

$$= \frac{0.01249}{0.01083}$$

$$= 1.153$$

$$df_1 = n_B - 1 = 8$$

$$df_2 = n_A - 1 = 8$$

Thus,

$$f_{0.05, 8, 8} = 3.44$$

$$f_{0.95, 8, 8} = \frac{1}{3.44} = 0.29$$

Critical region: $f < 0.29$ or $f > 3.44$

As $0.29 < 1.153 < 3.44$, we do not reject H_0 .

Thus, not enough evidence to conclude different variability, as they appear to have equal variability.

Qno.3:

$$H_0: \mu_1 = \mu_2 \text{ OR } \mu_d = 0$$

$$H_1: \mu_1 > \mu_2 \text{ OR } \mu_d > 0$$

$$\alpha = 0.05$$

Radial	Belted	Difference (d)
4.2	4.1	0.1
4.7	4.9	-0.2
6.6	6.2	0.4
7.0	6.9	0.1
6.7	6.8	-0.1
4.5	4.4	0.1
5.7	5.7	0
6.0	5.8	0.2
7.4	6.9	0.5
4.9	4.7	0.2
6.1	6.0	0.1
5.2	4.9	0.3
		$\Sigma d = 1.7$

$$n = 12$$

$$\bar{d} = 1.7/12 = 0.1417$$

$$s_d = \sqrt{\frac{\Sigma (d_i - \bar{d})^2}{n-1}} = 0.1975$$

$$\text{Critical region: } t_{0.05, 11} < t$$

$$t > 1.796$$

$$t = \frac{\bar{d} - \mu_d}{s_d/\sqrt{n}}$$

$$= \frac{0.1417}{0.1975/\sqrt{12}}$$

$$= \frac{0.1417}{0.0570} = 2.485$$

As $2.485 > 1.796$, we reject the H_0 .

Qno. 4:

H_0 : Type of crime is independent of district

H_1 : Type of crime is dependent on district.

$$\alpha = 0.01$$

$$df = (\text{rows} - 1) \times (\text{columns} - 1)$$

$$= 3 \times 3 = 9$$

Critical region: $\chi^2 > \chi^2_{0.01, 9}$

$$\chi^2 > 21.667$$

Row Totals:

$$R_1: \text{District 1} = 749$$

$$R_2: \text{District 2} = 1527$$

$$R_3: \text{District 3} = 919$$

$$R_4: \text{District 4} = 864$$

Column Totals:

$$C_1: \text{Assault} = 1010$$

$$C_2: \text{Burglary} = 682$$

$$C_3: \text{Larceny} = 2295$$

$$C_4: \text{Homicide} = 72$$

$$T: \text{Grand Total} = 4059$$

$$\text{Expected frequency} = \bar{E}_i = \frac{(R_i \times C_i)}{T}$$

$O_i | \bar{E}_i$ table

District	Assault	Burglary	Larceny	Homicide
1	162 186.4	118 125.8	451 423.5	18 13.3
2	310 380	196 256.6	996 863.4	25 27.1
3	258 228.7	193 154.4	458 519.6	10 16.3
4	280 214.9	175 145.2	390 488.5	19 15.3

$$\chi^2 = \frac{\sum (O - E)^2}{E}$$

District	Assault	Burglary	Larceny	Homicide	
1	3.19	0.49	1.79	1.66	
2	12.89	14.31	20.36	0.16	
3	3.75	9.65	7.30	2.43	
4	19.72	6.12	19.86	0.89	$\chi^2 =$
	39.55	30.57	49.31	5.14	124.57

As $124.57 > 21.667$, we reject H_0 . Thus, type of crime is dependent on district.

Qno. 5:

H_0 : Digits follow discrete uniform PMF

H_1 : Digits do not follow discrete uniform PMF

$$\alpha = 0.05$$

$$N = 114$$

For uniform discrete PMF, each digit should be equally frequent to all others.

$$E = \frac{114}{10} = 11.4$$

Digit	Observed	Expected	$(O-E)^2/E$
0	17	11.4	2.751
1	16	11.4	1.856
2	13	11.4	0.225
3	10	11.4	0.172
4	8	11.4	1.014
5	12	11.4	0.032
6	12	11.4	0.032
7	12	11.4	0.032
8	7	11.4	1.698
9	7	11.4	1.698
			$\Sigma = 9.51$

Critical region: $\chi^2 > \chi^2_{0.05, 9}$
 $\chi^2 > 16.919$

As $9.51 < 16.919$, we do not reject H_0 .